

OBJECTIVE

Backend-focused software engineer with 3 years of experience designing automation platforms and building scalable web applications using Python. Seeking a backend engineering role where I can contribute to API development, cloud-native systems, and AI-powered applications while continuing to grow in distributed systems and software architecture.

HIGHLIGHTS

- **Backend Frameworks & APIs:** Skilled in developing robust server-side logic and RESTful APIs using Django, **Django REST Framework (DRF)**, and Flask. Experienced in managing Object-Relational Mapping (ORM) and implementing secure authentication protocols (e.g., JWT, OAuth 2.0) for seamless third-party integrations.
 - **Core Python & Problem Solving:** Strong command of Python and core Data Structures and Algorithms. Capable of translating complex business logic into efficient code, utilizing Regular Expressions (Regex) and data manipulation techniques for advanced text processing, validation, and algorithmic challenges.
 - **Frontend Development:** Proficient in building interactive, responsive Single-Page Applications (SPAs) using **React**. Capable of utilizing React Hooks, state management, and data visualization libraries to translate complex backend data into intuitive user interfaces.
 - **Machine Learning & AI:** Solid academic foundation in Artificial Intelligence principles and supervised/unsupervised learning algorithms. Possess a solid understanding of deep learning frameworks (PyTorch) and the Transformer architecture.
 - **AI-Augmented Development:** Skilled in leveraging AI coding assistants (e.g., GitHub Copilot, Cursor) to accelerate feature implementation, assist with code refactoring, maintaining high efficiency and code quality.
 - **Database & Performance:** Skilled in managing relational data with **PostgreSQL**, including query optimization via database indexing. Proficient in utilizing Redis caching to reduce latency and enhance system response times under high-concurrency traffic.
 - **DevOps & Containerization:** Proficient in using Docker for environment isolation and streamlining deployment workflows, ensuring consistency across development and production stages.
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PROFESSIONAL EXPERIENCE

Software Engineer (Automation & Testing Infrastructure)

August 2021 – August 2024

Shanghai SAIC Mobility Technology and Service Co., Ltd | Shanghai, China

A leading enterprise-level mobility platform under the Fortune Global 500 SAIC Motor Group

- Managed and optimized testing operations for SAIC Mobility, supporting millions of daily active users and high-concurrency service requests.
- Led the design and development of a Python-based automation platform, defining system architecture and core modules, enabling scalable test development and reducing manual testing effort by 40%.
- Owned the framework architecture, including modular design and extensibility strategy, ensuring maintainability during rapid product iterations.
- Designed and implemented an API-driven configuration injection system to control runtime environments, eliminating test flakiness caused by inconsistent system states.
- Built and maintained a data sandbox infrastructure to simulate complex user scenarios with isolated datasets, enabling deterministic validation of edge-case workflows.
- Drove cross-team integration with backend, frontend, and DevOps teams to stabilize environments and support CI/CD pipelines in high-concurrency systems.
- Mentored engineers and conducted code reviews, establishing coding standards and improving overall engineering quality across the team.

PROJECT EXPERIENCE

Lace Up | Full-Stack Running Analytics Platform

<https://andrewlu3335.github.io/>

Independent full-stack Developer

- Tech Stack: React, Django REST Framework, PostgreSQL (Neon over SSL), Redis, Celery Beat, Docker, AWS ECS Fargate, Amazon ECR, Amazon S3, CloudFront, CloudWatch Logs, IAM, Strava API.
 - Designed the application architecture following a decoupled frontend-backend model, separating React client delivery from Django API services to improve maintainability and deployment flexibility.
 - Architected and deployed a cloud-native full-stack application on AWS using ECS Fargate, ECR, S3, and CloudFront for scalable hosting and content delivery.
 - Implemented secure database connectivity using Neon PostgreSQL over SSL and environment-based secret management.
 - Built asynchronous data synchronization pipelines using Celery Beat and Redis to periodically synchronize Strava activities and enrich workout records with weather data.
 - Configured CloudWatch logging and IAM role-based access control to improve observability and operational security.
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EDUCATION

Shanghai Dianji University— Shanghai, China

June 2021

Bachelor of Engineering in Communication Engineering

Scholarship Recipient

Laurentian University— Sudbury, Ontario, Canada

May 2026

Master of Science in Computational Sciences

Relevant Coursework:

Artificial Intelligence,

Machine Learning & Deep Learning,

Applied Cryptography,

Intro to Cloud Technologies

Cumulative Average: 9.40 / 10.0

TECHNICAL SKILLS

Coding Languages: Python, SQL.

Backend: Django, Django REST Framework (DRF), REST API Design, JWT, OAuth 2.0

Databases & Caching: PostgreSQL, Redis

Cloud & DevOps: AWS ECS Fargate, Amazon ECR, Amazon S3, Amazon CloudFront, CloudWatch, IAM, Docker, Git

Frontend: React

AI Tools: Copilot, Cursor
